

## Class 6

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  - Midterm - Due Mon Oct 5 11:59:59 PM EDT 2026

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- Robustness of peer-to-peer systems
  - *Robustness* - addresses the resiliency and integrity of the system
  - *Churn* - the opposite of efficiency
  - *Gossip* - Chapter 3 or 4
- Edge Computing
  - *Provisioning* - for cloud based hosting environments
  - *Allocation* - edge application may need some allocation routines
    - \* Tools to support this such as **k8s**
- Virtual machines versus container technology
  - Appears to be a movement between virtual machines -> containers
  - Main reasoning - containers can allow for performance gains over virtual machines
- Scalability of publish-subscribe systems
  - A *content-based* publish-subscribe system scales at  $O(n \cdot m)$ 
    - \* Let  $n$  be the number of subscribers
    - \* Let  $m$  be the number of topics
  - Customers can subscribe to a matching property
  - Topic-based vs Channel-based
  - They want to throw in anonymity
- Scalability of blockchain systems

Each topic is for 2 points, the midterm is 10-points total. Each one is worth ~3 paragraphs (each paragraph being ~3 sentences) to complete. Maybe run at a page for the upper-bound. <sup>1</sup>

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<sup>1</sup>**For Dan** no more than 1 page per each of the 5 topics.